

ARCHITECTURE OF INTERNET-SCALE NAMING

Advanced Systems Architecture
Distributed Data Structures

LEARNING OBJECTIVES



Analyze the 3-layer architecture of internet-scale naming spaces.



Compare structured path resolution with attribute-based discovery.



Deconstruct decentralized multi-dimensional data indexing.



Understand the application of Hilbert curves in Distributed Hash Tables (DHTs).

Master the transition from physical paths to mathematical coordinate indexing.

THE NECESSITY OF DISTRIBUTION

“A single global directory is technically unfeasible (Cheriton & Mann, 1989).”

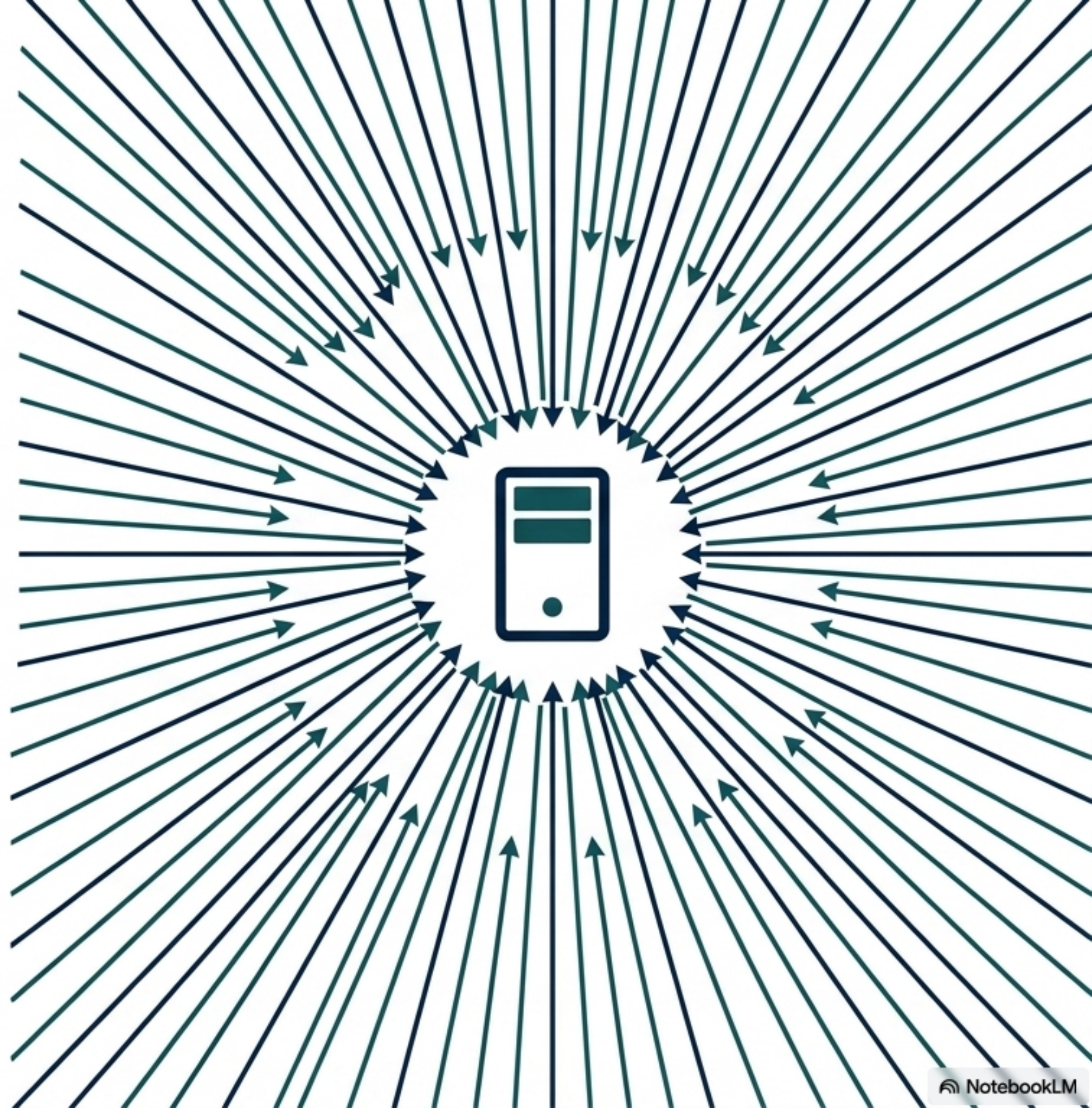
GEOGRAPHICAL LATENCY:

Centralization punishes distant users across wide-area networks.

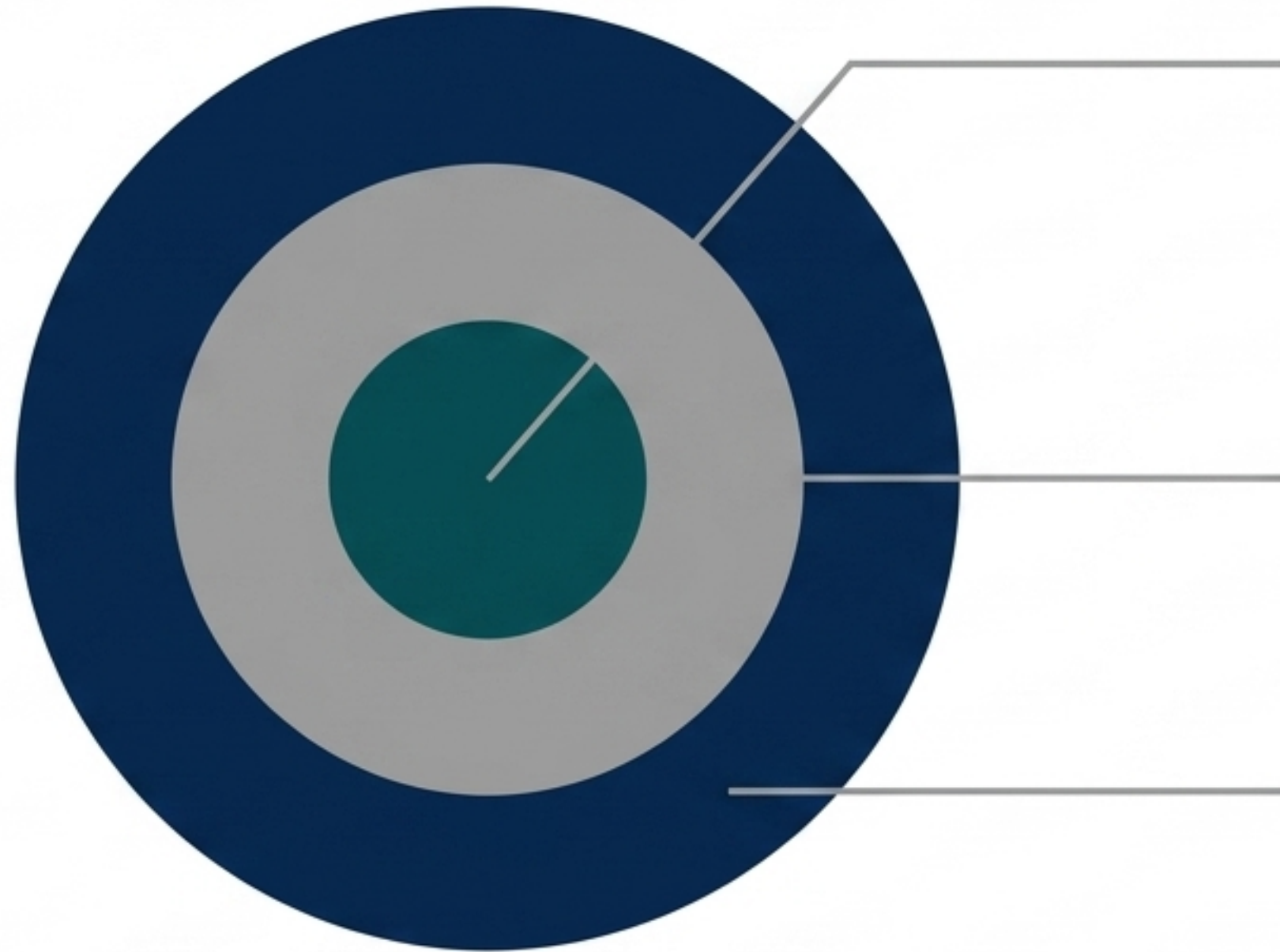
ENTITY VOLUME:

Single nodes lack the memory for billion-scale traffic.

We must partition name spaces into logical layers based on stability and scale.



PARTITIONING THE NAME SPACE



MANAGERIAL LAYER

The dynamic edge. Individual user hosts, local network nodes, and shared files.

ADMINISTRATIONAL LAYER

The organization (e.g., a university). Balances internal operational needs with external visibility.

GLOBAL LAYER

The foundation (e.g., .com, .edu). Highly stable, rarely changes.

Rules for replication and caching shift at every layer based on volatility.

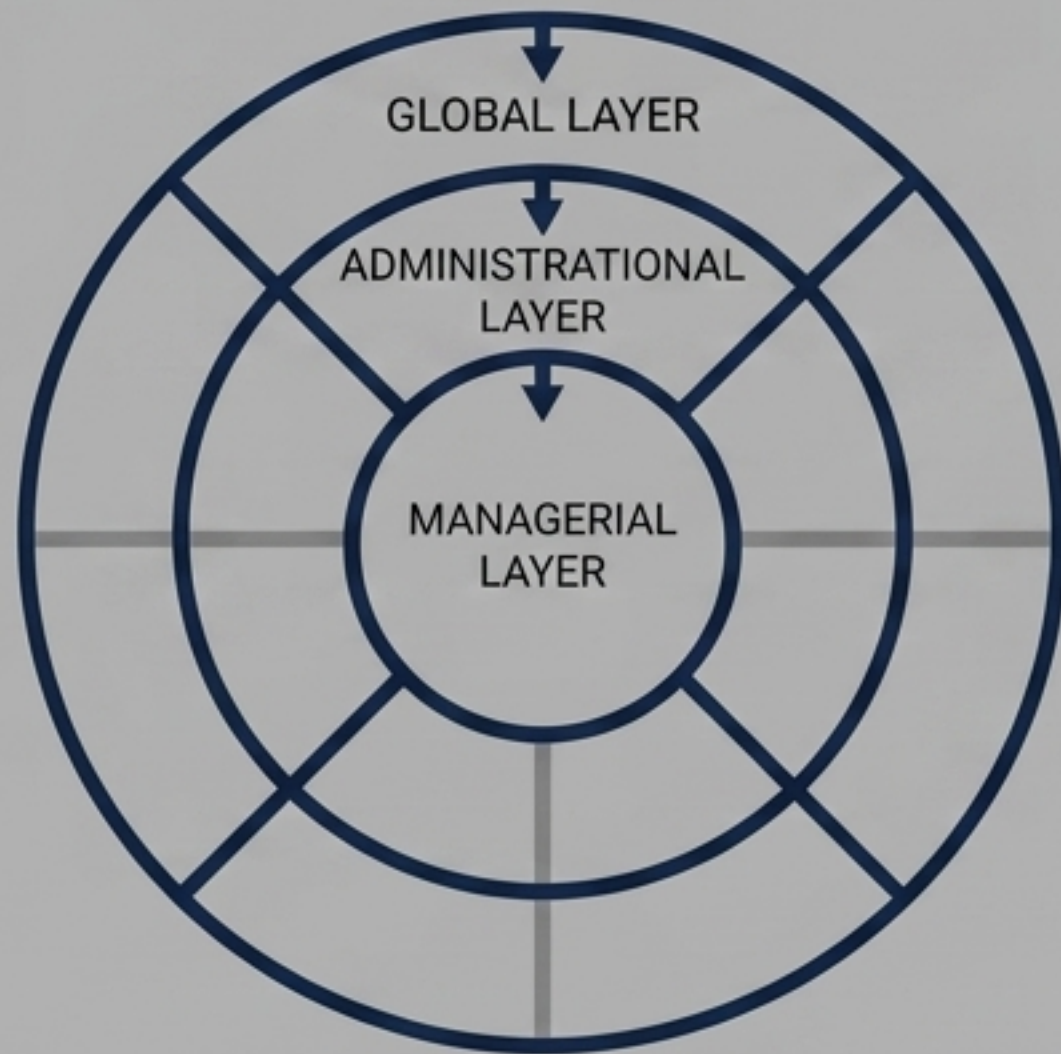
THE CONSISTENCY VERSUS PERFORMANCE TRADE-OFF

FEATURE	GLOBAL LAYER	ADMINISTRATIONAL LAYER	MANAGERIAL LAYER
Scale	Worldwide	Organization	Department/User
Node Count	Few	Many	Vast numbers
Lookups	Seconds	Milliseconds	Immediate
Updates	Lazy	Immediate	Immediate
Caching	Highly Effective	Effective	Ineffective

Global layers favor availability via lazy updates; Managerial layers demand immediate consistency.

DISTINGUISHING LOGICAL LAYERS FROM PHYSICAL ZONES

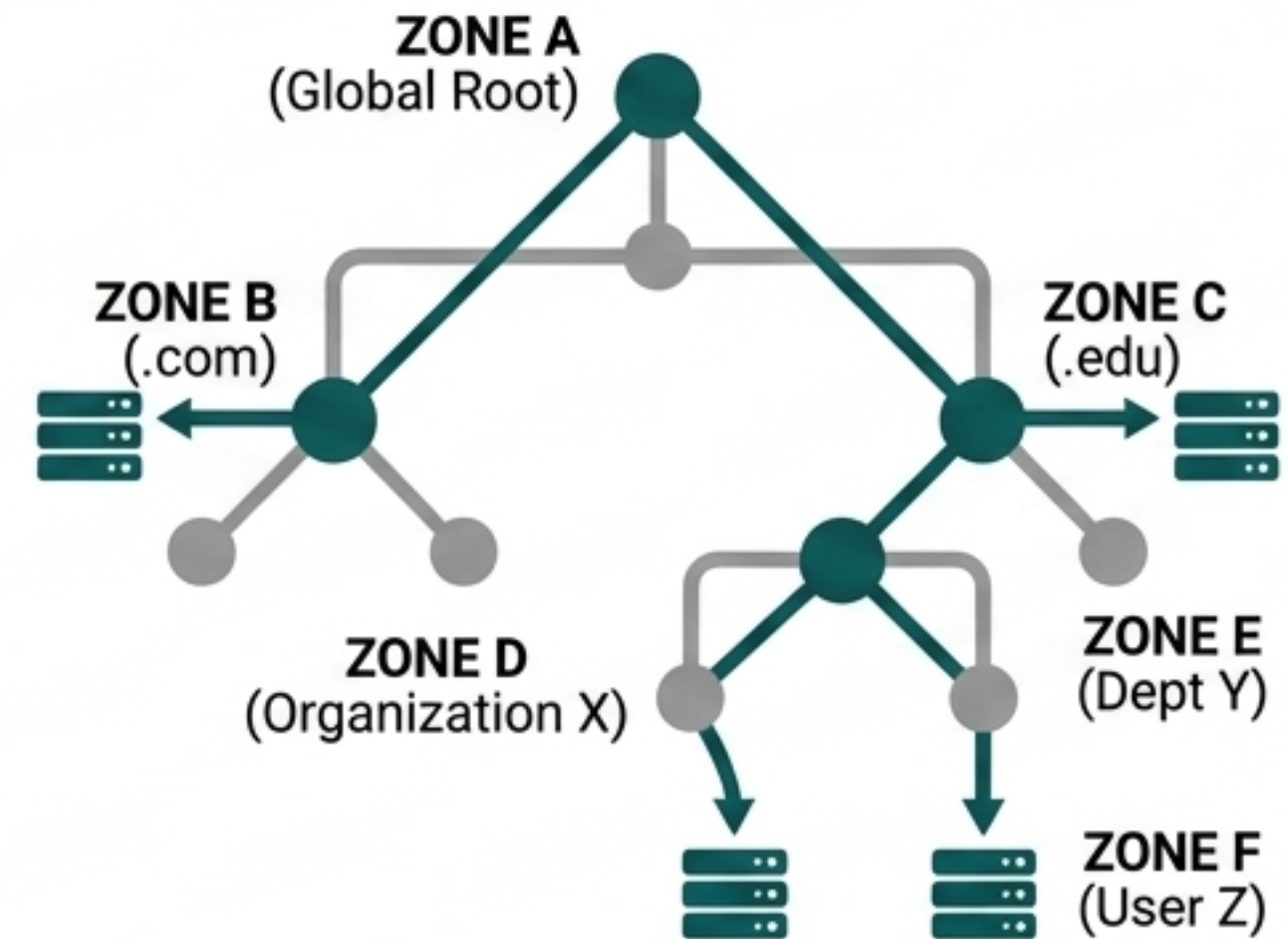
Logical Hierarchy (Layers)



Conceptual categorization of data based strictly on its stability and scale.

PHYSICAL IMPLEMENTATION (ZONES)

Specific, non-overlapping parts of the namespace managed by a single physical name server (e.g., DNS Zones).



Layers describe the **nature** of the data; **Zones** describe the infrastructure hosting it.

The takeaway text spans the bottom across both colored sections in a separate white block.

THE EVOLUTION FROM PATHS TO ATTRIBUTES

STRUCTURED NAMING (DNS)

Question: Where is it?

Identifier: Unique path or string.

Nature: Static, highly location-independent.

Result: Locates a single, specific entity.

ATTRIBUTE-BASED NAMING (DIRECTORY SERVICES)

Question: What is it?

Identifier: (Attribute, Value) pairs.

Nature: Dynamic, human-friendly descriptors.

Result: Discovers multiple matching entities.

Modern distributed systems demand resource discovery, fundamentally breaking rigid 1D path lookups.

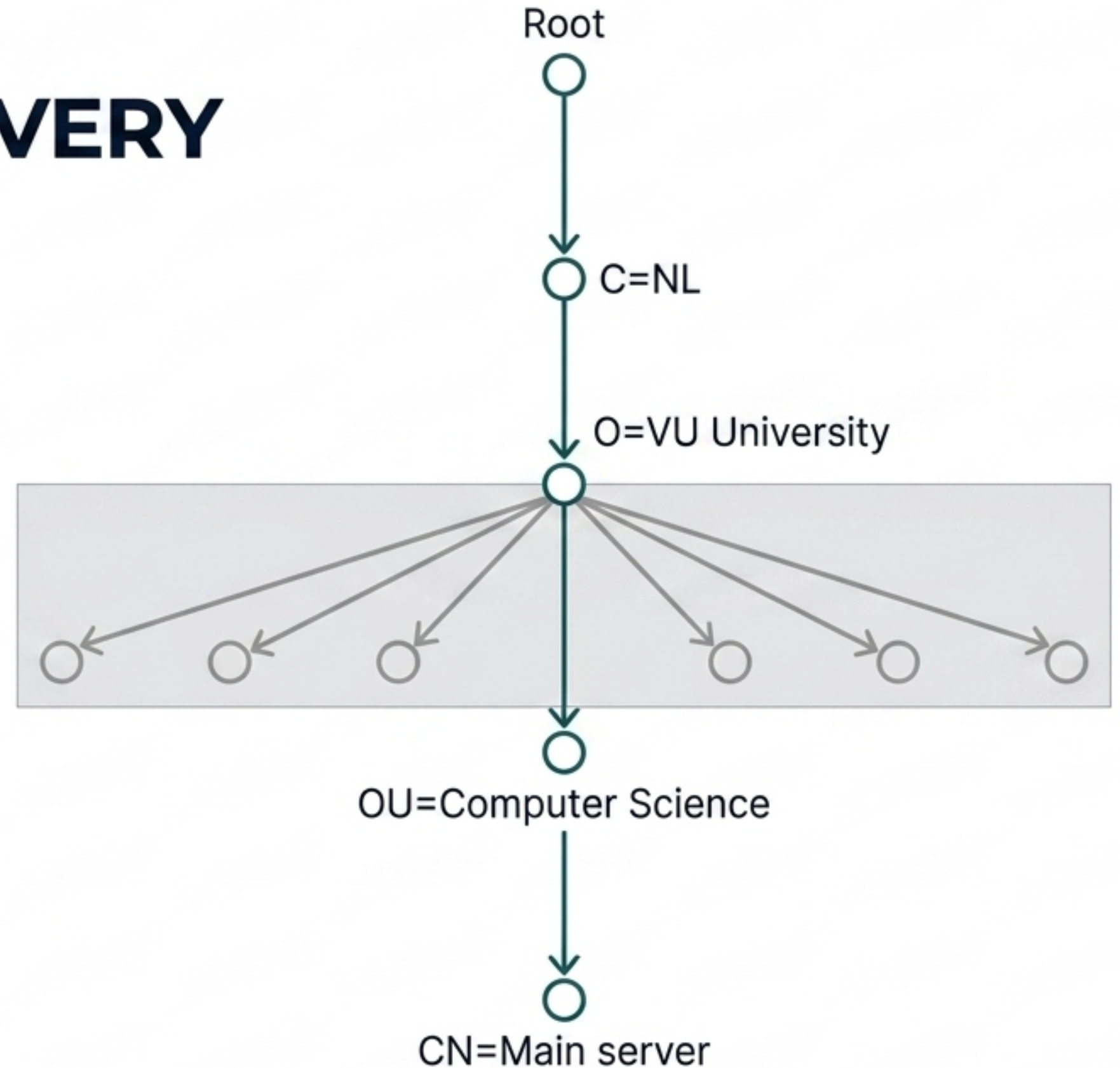
THE SEARCH COST OF HIERARCHICAL DISCOVERY

Organizes entries into a Directory Information Tree (DIT).

Navigates via sequential **Relative Distinguished Names** (RDNs).

The Search Cost:

- To find all main servers at a university, LDAP cannot use a direct path; it must exhaustively search every departmental leaf node.
- Hierarchical searching forces expensive top-down coordination across the network.



THE BOTTLENECKS OF DECENTRALIZED INDEXING

Assigning each attribute to independent distributed servers creates severe **architectural failures**:



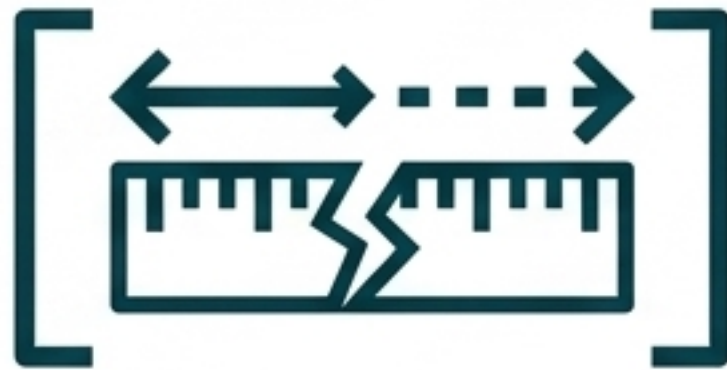
1. COMMUNICATION OVERLOAD

Querying k attributes requires contacting k distinct servers across the WAN.



2. CLIENT PROCESSING DRAIN

Filtering millions of unfiltered keys (e.g., all users named Smith) is pushed entirely to the client.



3. RANGE QUERY FAILURE

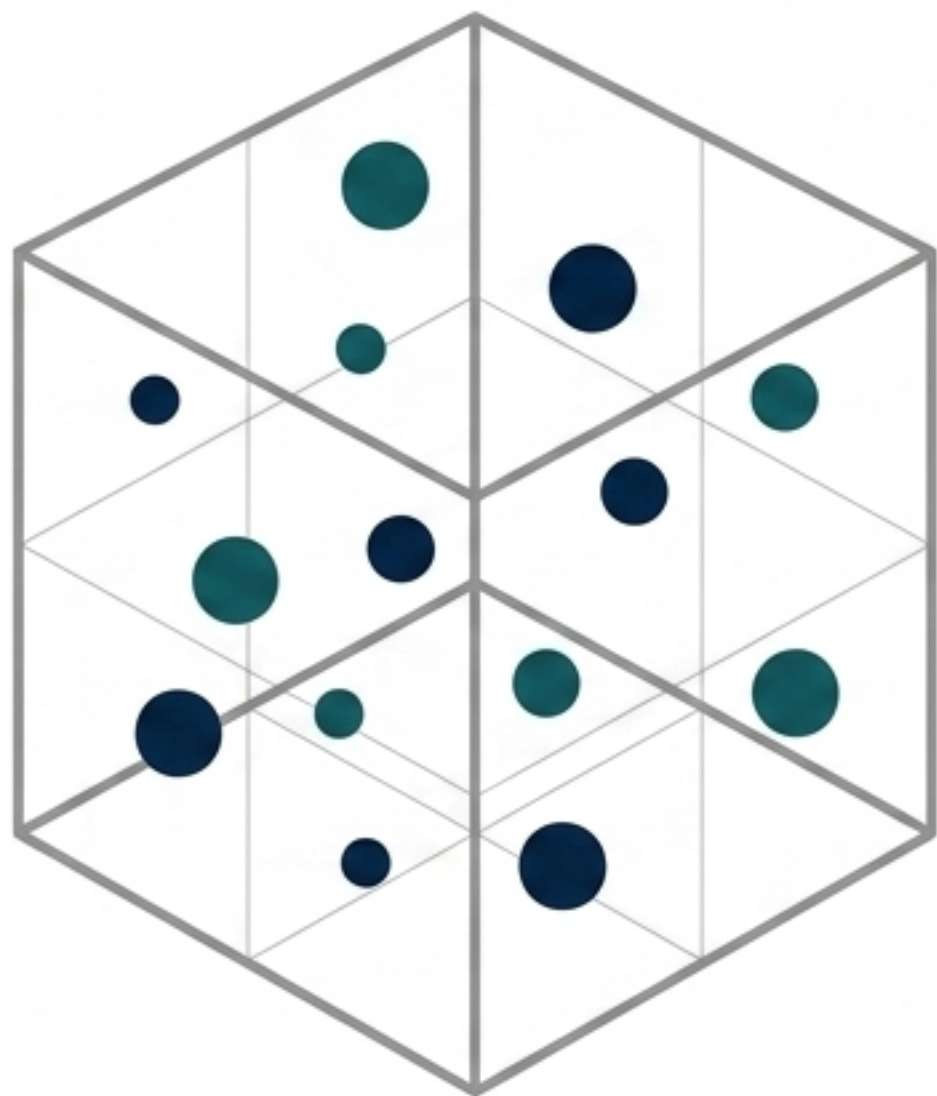
Direct hashing excels at exact matches but fails at descriptors like Price = \$1000–\$2500.

Multi-dimensional data requires a mathematical, non-hierarchical routing approach.

FLATTENING DIMENSIONS WITH SPACE-FILLING CURVES

The Goal:

- Flatten multi-dimensional attribute space (N-dimensions) into a linear index (1D line).



The Mechanism:

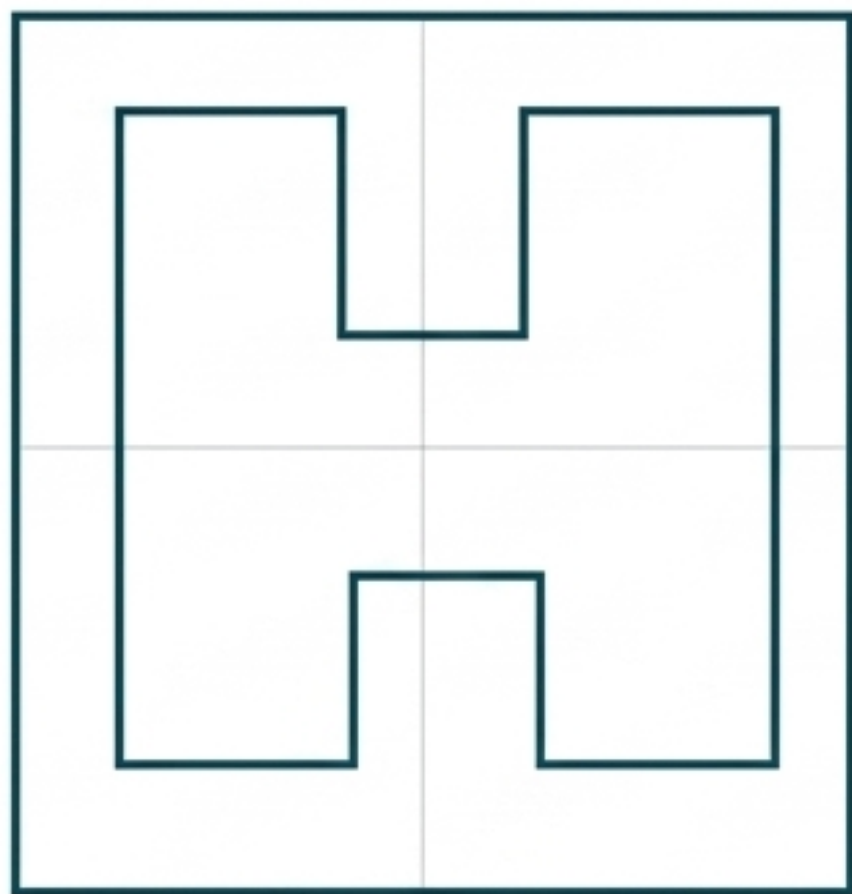
- Mathematical functions map complex data coordinates to a continuous linear path without losing their relationship.



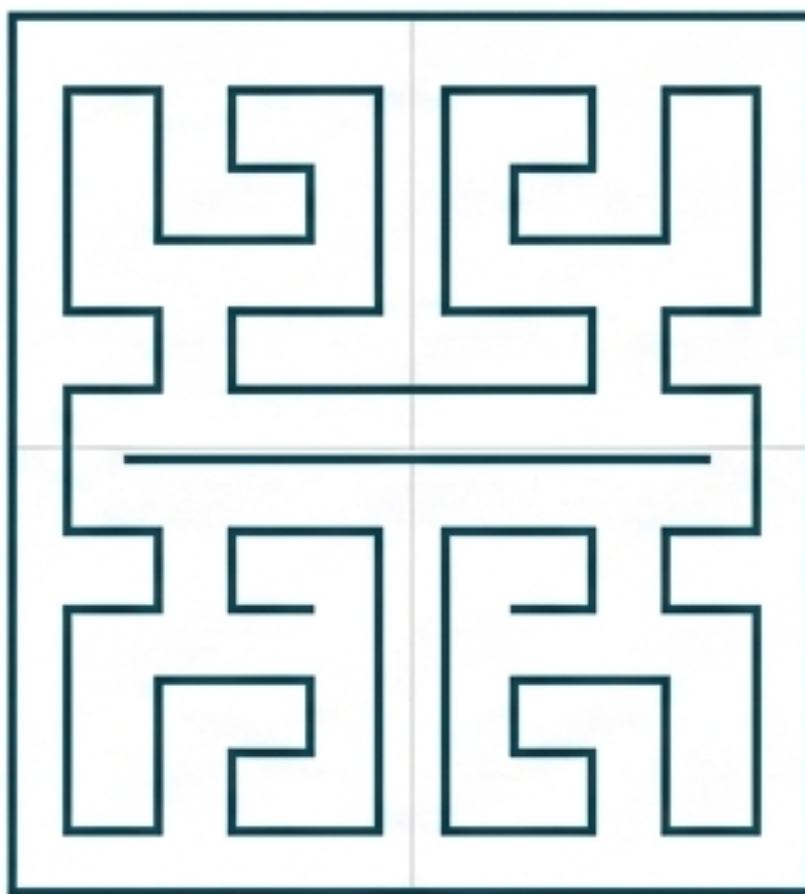
The Result:

- Enables partitioning across a network while preserving data locality.
- Related multi-dimensional attributes are co-located on the same server, eliminating massive cross-network queries.

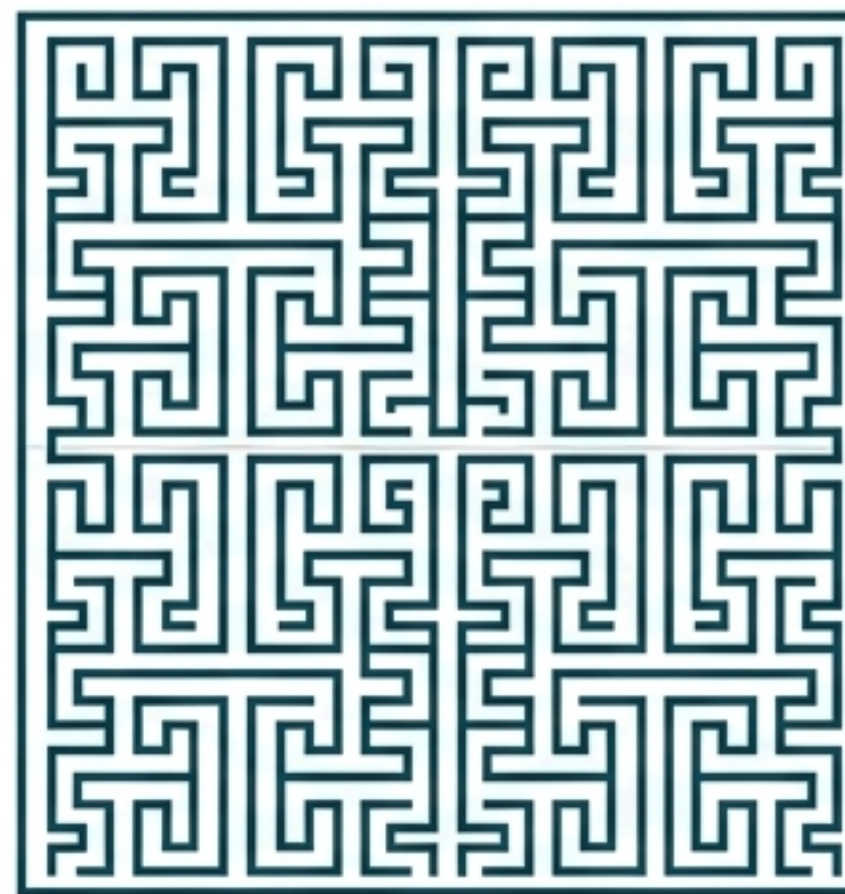
PRESERVING LOCALITY VIA HILBERT CURVES



ORDER 1



ORDER 2



ORDER 4

Recursively divides N-dimensional space into quadrants.

Standard real-world systems use **Order 32 or 64** (2^{64} total indices).

Connects **all quadrants** with a **single continuous line**.



THE ARCHITECT'S WARNING:

Proximity in multi-dimensional space usually preserves locality on the curve, but not always. Outliers exist.

Complex **geometric proximity** is translated into **simple linear distance**.

MAPPING ATTRIBUTES TO STORAGE

1. NORMALIZATION

Map all attribute values to the mathematical interval $[0, 1)$.



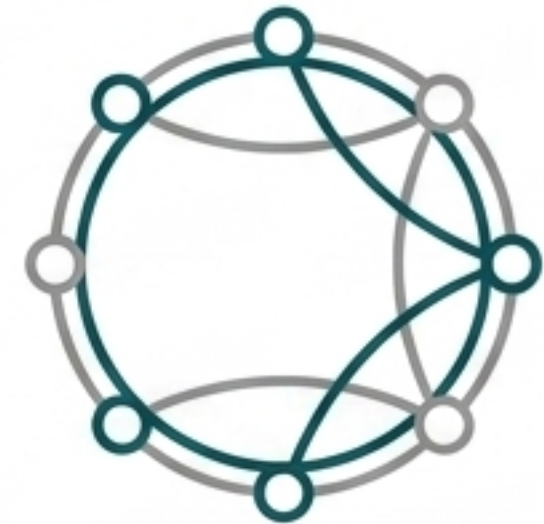
2. INDEXING

Assign these coordinates to an index on a high-order Hilbert curve.



3. STORAGE

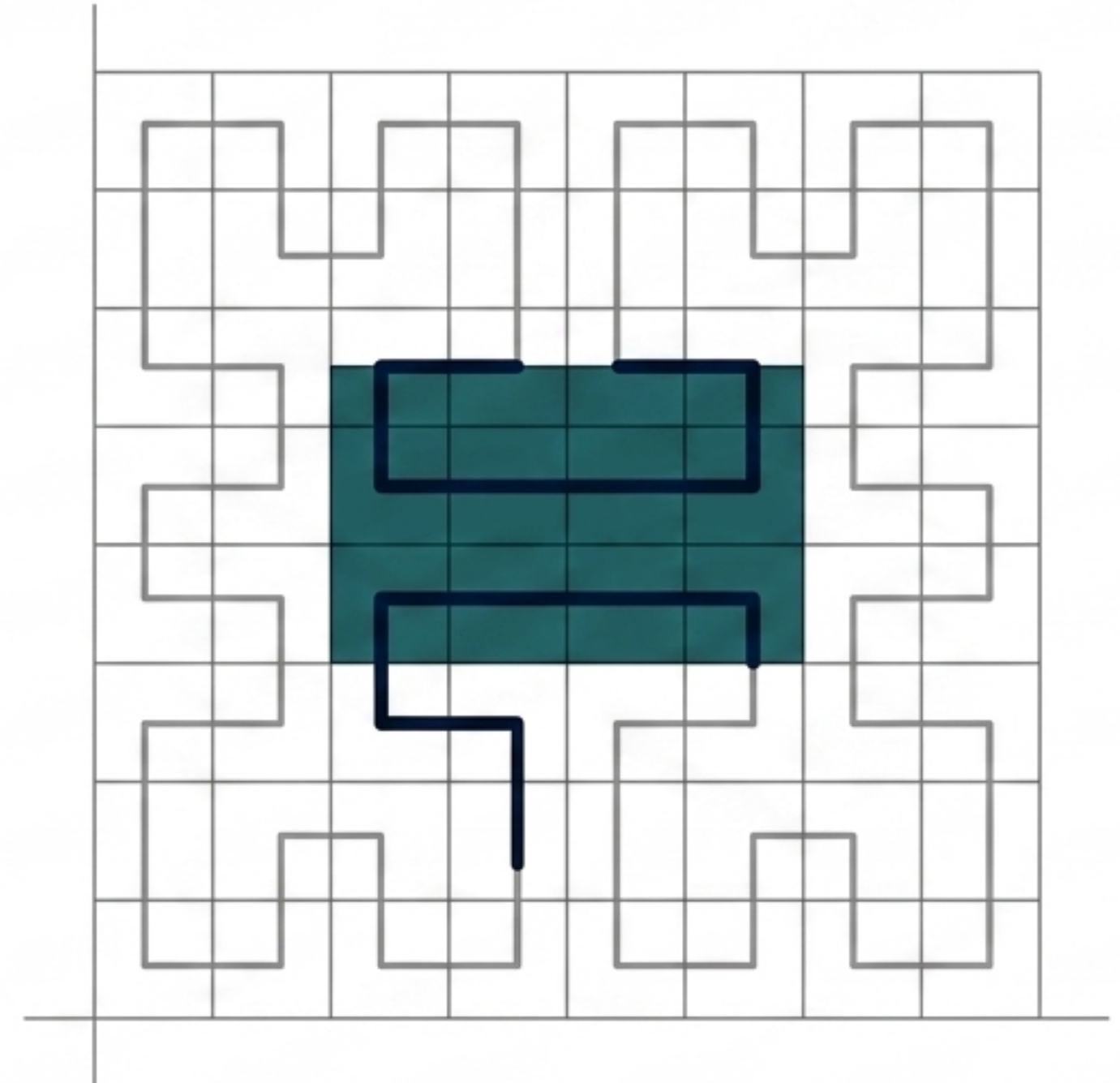
Map the resulting 1D indices directly to a Distributed Hash Table (DHT) identifier space, such as a Chord ring.



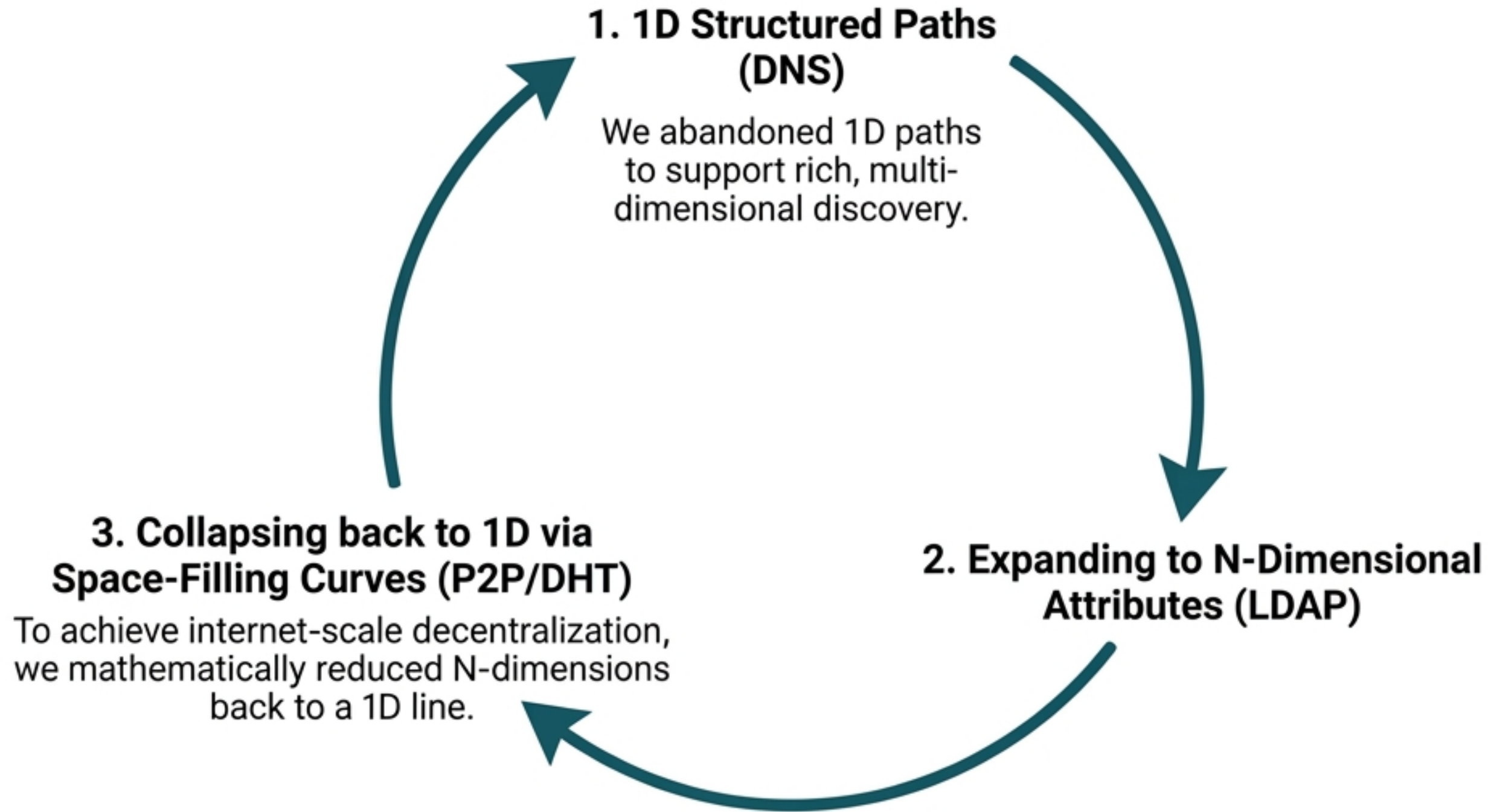
We perfectly align the mathematical Hilbert index space with the DHT's physical network space.

RESOLVING MULTI-DIMENSIONAL RANGE QUERIES

- Range queries (e.g., price between 0.5 and 0.7) define a conceptual rectangular region in space.
- The algorithm calculates which specific segments of the 1D Hilbert curve intersect that rectangle.
- Efficiency: Queries are routed only to the specific network nodes responsible for those intersecting curve segments.
- Mathematical intersection replaces exhaustive network searching.



THE DIMENSIONALITY CYCLE



Scalability is fundamentally an exercise in dimensional reduction.

ARCHITECTURE SUMMARY

Distribution is Mandatory: Single-node limits force logical partitioning. DNS scales via layered hierarchies (Global to Managerial).

Discovery Requires Attributes: Structured searching limits usability. LDAP provides N-dimensional discovery but introduces top-down network bottlenecks.

Math Enables P2P Scale: Hilbert curves map complex, multi-attribute data to 1D Distributed Hash Tables, resolving latency and query costs.

Internet-scale systems survive by ruthlessly isolating local volatility and mathematically compressing dimensions.